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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***

Code: 105602 | ***Subject: Machine Learning*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option for the following questions (Answer any seven): **[2 x 7 = 14]**

(a) Gradient Descent is an example of
- (i) Evolutionary algorithm
- (ii) Sampling algorithm
- (iii) Optimization algorithm
- (iv) Clustering algorithm

(b) Which regression technique performs feature selection?
- (i) Ridge Regression
- (ii) Linear Regression
- (iii) LASSO
- (iv) Polynomial Regression

(c) PCA chooses components that
- (i) Maximize variance
- (ii) Maximize covariance
- (iii) Minimize variance
- (iv) Increase dimensionality

(d) In Bayesian Learning, a conjugate prior is chosen because it
- (i) simplifies posterior calculation
- (ii) eliminates noise
- (iii) improves accuracy
- (iv) selects features

(e) Logistic Regression uses
- (i) Linear activation
- (ii) ReLU activation
- (iii) Sigmoid activation
- (iv) hyperbolic tangent

(f) Perceptron works only on
- (i) Linear separable data
- (ii) High-dimensional data
- (iii) Non-linear data
- (iv) Unsupervised data

(g) Bagging mainly helps in reducing
- (i) Bias
- (ii) Variance
- (iii) Noise
- (iv) Over-sampling

(h) K-medoids is more robust to
- (i) Large data
- (ii) Gaussian distribution
- (iii) Outliers
- (iv) High variance

(i) Bayesian networks represent
- (i) Undirected dependencies
- (ii) Fully connected graph
- (iii) Directed acyclic graph
- (iv) Weighted graph

(j) K-means uses
- (i) Manhattan distance
- (ii) Medoids
- (iii) Centroids
- (iv) Probabilities

Q.2

(a) Define hypothesis space and inductive bias with examples. [7]

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(b) Explain the difference between supervised, unsupervised, and reinforcement learning with real world applications. [7]

Q.3

(a) What is bias-variance trade-off? Describe K-fold cross-validation and its importance. [7]

(b) Draw the perceptron network with the notation. Derive an equation of gradient descent rule to minimize the error. [7]

Q.4

(a) Explain PCA with derivation of eigen decomposition and dimensionality reduction. [7]

(b) Compare ML, MAP and Bayes estimators using an example. [7]

Q.5

(a) Describe ANN architecture and derive back propagation equations. [7]

(b) Explain Decision Tree learning using Gini Index & Information Gain. [7]

Q.6

(a) Explain ensemble learning. How do Bagging and Boosting reduce error? [7]

(b) Discuss hypothesis-testing and its role in ML model comparison. [7]

Q.7

(a) Compare K-means, K-medoids, and spectral clustering in detail. [7]

(b) Derive the Expectation-Maximization algorithm step-by-step. [7]

Q.8

(a) Explain reinforcement learning with policy, reward, and value functions. [7]

(b) Describe Bayesian Networks and inference techniques. [7]

Q.9 Write short notes on any two of the following: **[2 x 7 = 14]**

(a) Support Vector Machine

(b) Linear Regression

(c) Gaussian Mixture Model

(d) AdaBoost algorithm

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